FEATURE BASED PATH-PLANNING

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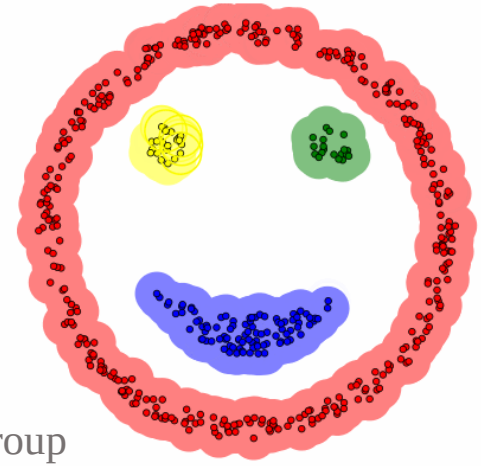
INTRODUCTION

- Research Background
  - In INS/GPS integrated system, if the GPS signal is unavailable, the estimation diverge
    - As integration errors accumulate over time, the estimation diverge
  - Fusing with another sensors(such as camera, LiDAR, or radar)
    - The INS/Camera system is cost-effective
  - INS/Camera system is highly dependent on feature points
    - Feature point have unique pattern

- Research Objective
  - Military areas and mountainous terrain have sparse feature points
  - Path planning via feature points

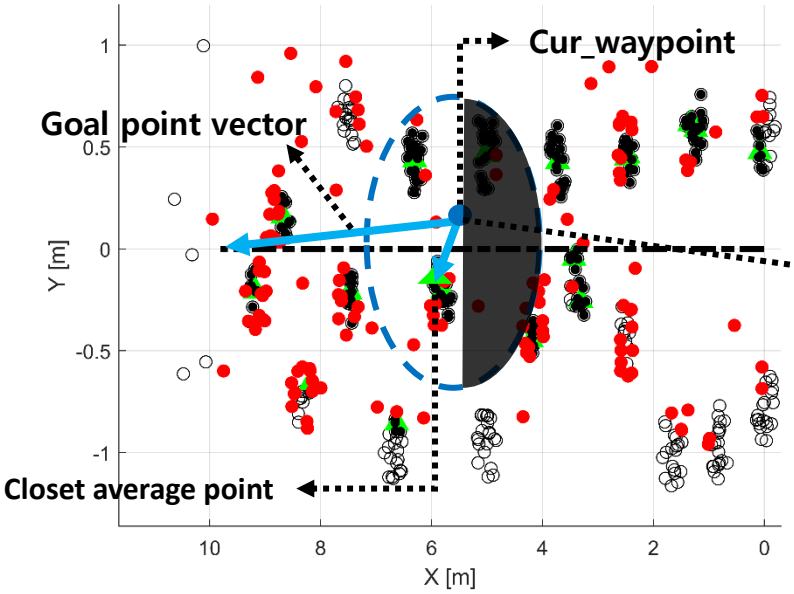
PATH PLANNING (1 / 5) - DBSCAN

- DBSCAN
  - Density Based Spatial Clustering of Applications with Noise



- Initial setup
  - Epsilon
    - Maximum distance between points
    - Points withing a certain radius are considered to be in the same group
  - Minimum Points
    - Minimum number of neighboring points required for group formation
- Group formation process
  - Select point
  - Calculate the number of neighboring points within the radius
  - If the number of neighboring points exceeds the threshold, group formation
  - Repeat the process

PATH PLANNING (2 / 5) - ICROS



- Noise
  - Points that do not belong to any group
- Average point
  - Indicates area with a high distribution of feature points

- Feature point vector
  - Use to traverse areas with a high distribution of feature points
  - The vector pointing from the current waypoint to the closet average point
- Goal point vector
  - Used to maintain direction toward the target point
  - The vector pointing from the current waypoint to the target point

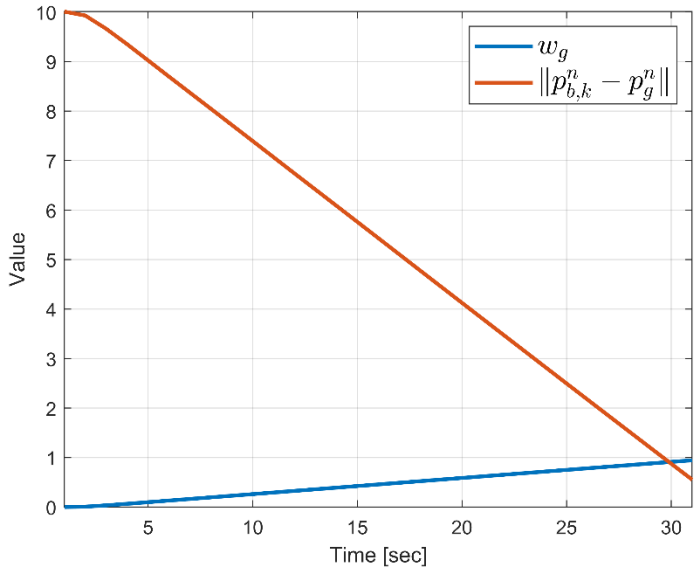
PATH PLANNING(3 / 4) – ICROS

- Weight calculation

$$w_f = \begin{cases} \dfrac{\|\boldsymbol{p}_{b,k}^n - \boldsymbol{p}_g^n\|}{\|\boldsymbol{p}_{b,0}^n - \boldsymbol{p}_g^n\|} & \text{feature in FOV} \\ 0 & \text{else} \end{cases}$$

$$w_f + w_g = 1$$

$\boldsymbol{p}_{b,k}^n$	Current waypoint w.r.t navigation frame
$\boldsymbol{p}_g^n$	Goal point
$\boldsymbol{p}_{b,0}^n$	Start point
w_f	The weight of the feature point vector
w_g	The weight of the goal point vector

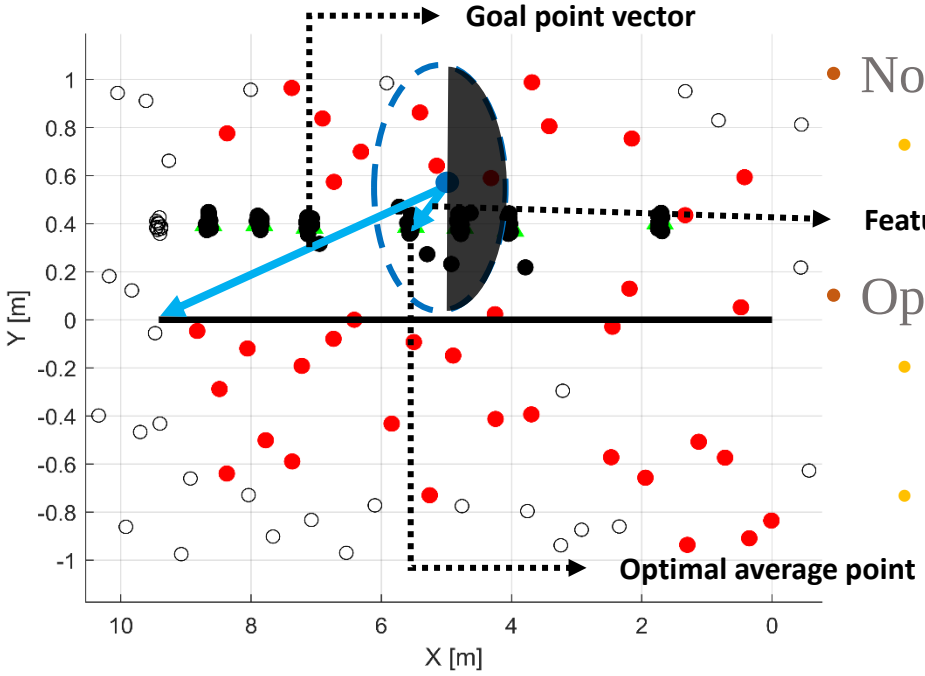


- Waypoint calculation

$$\boldsymbol{p}_{b,k+1}^n = \boldsymbol{p}_{b,k}^n + \frac{w_f\, C_b^n\, p_{m,k}^{b,k} + w_g\, C_b^n\, p_g^{b,k}}{\left\| w_f\, C_b^n\, p_{m,k}^{b,k} + w_g\, C_b^n\, p_g^{b,k} \right\|_2}$$

$\boldsymbol{p}_{b,k+1}^n$	Next waypoint
$\boldsymbol{p}_{m,k}^n$	Closet average point

PATH PLANNING (4 / 5) - KCI



- Noise
  - Points that do not belong to any group
- Optimal average point
  - Minimize the sum of distance to all feature points within FOV
  - $\underset{m_i^n}{\operatorname{argmin}} \sum_j \left\| p_{bj}^n - m_i^n \right\|$

- Feature point vector
  - Use to traverse areas with a high distribution of feature points
  - The vector pointing from the current waypoint to the optimal average point
- Goal point vector
  - Used to maintain direction toward the target point
  - The vector pointing from the current waypoint to the target point

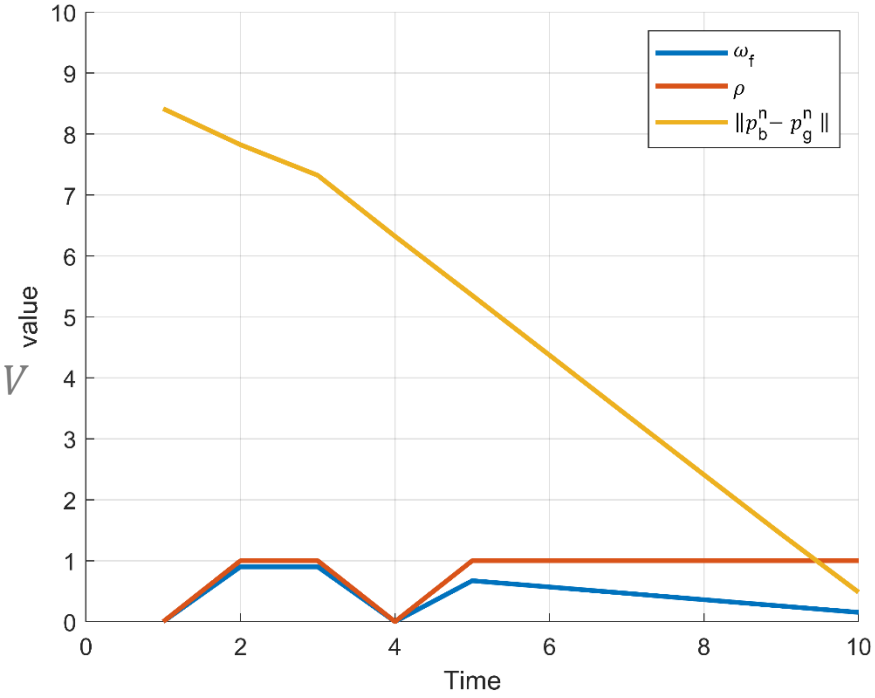
PATH PLANNING (5 / 5) - KCI

- Weight calculation

$$V_f = \sum_j \left\| p_{b,j}^n - m_i^n \right\|, \qquad \rho = \frac{N_f}{V_f}$$

$$\omega_f = \begin{cases} \dfrac{\left\| p_{b,k}^n - p_g^n \right\|}{\left\| p_{b,0}^n - p_g^n \right\|} \dfrac{1}{1 + e^{-\rho}}, & feature \in FOV \\ 0, & otherwise \end{cases}$$

$$w_f + w_g = 1$$



- Waypoint calculation

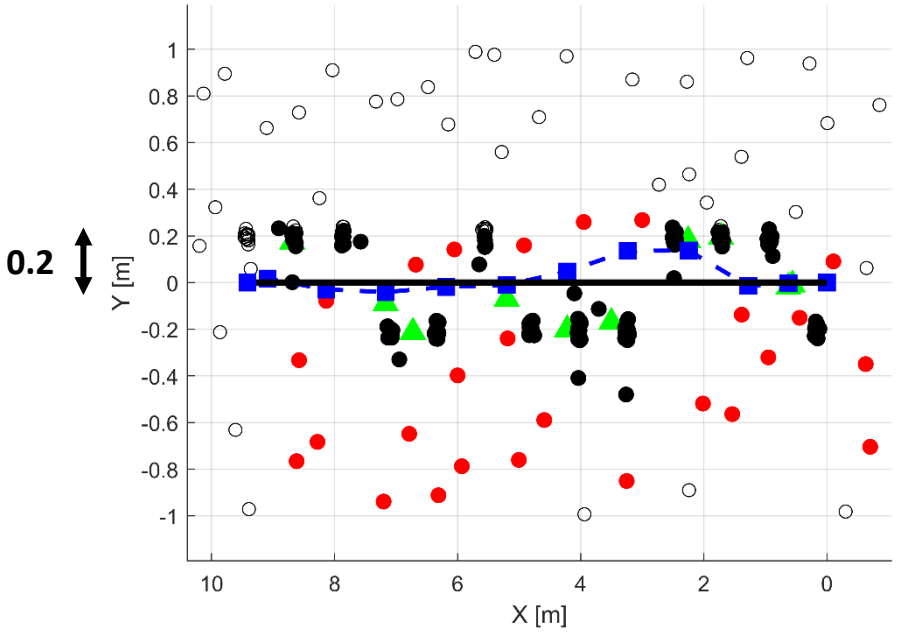
$$\boldsymbol{p}_{b,k+1}^n = \boldsymbol{p}_{b,k}^n + \frac{w_f\, C_b^n\, p_{m,k}^{b,k} + w_g\, C_b^n\, p_g^{b,k}}{\left\| w_f\, C_b^n\, p_{m,k}^{b,k} + w_g\, C_b^n\, p_g^{b,k} \right\|_2}$$

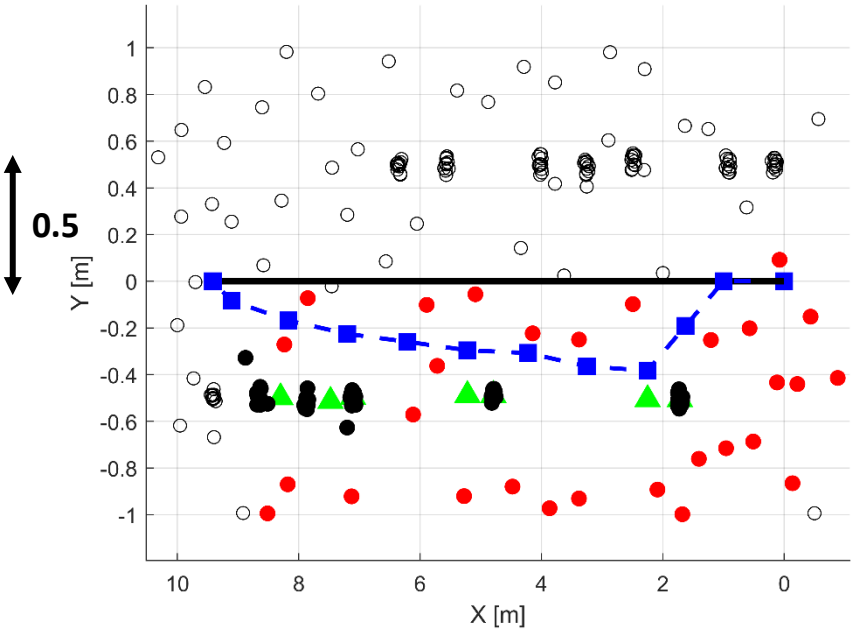
V_f	The distribution of feature points
$p_{b,j}^n$	The position of j-th feature points
m_i^n	Optimal average point
N_f	The number of feature points
ρ	The density of feature points

SIMULATION (1 / 3) - BOTH

- Simulation environment
  - Drone FOV with a radius of 0.8 meters
  - Epsilon : 0.3, minimum points : 5
  - 474 feature points

- Feature points considered in path planning
- Noise
- Optimal average point
- Feature points not considered in path planning
- LOS(Line of Sight)

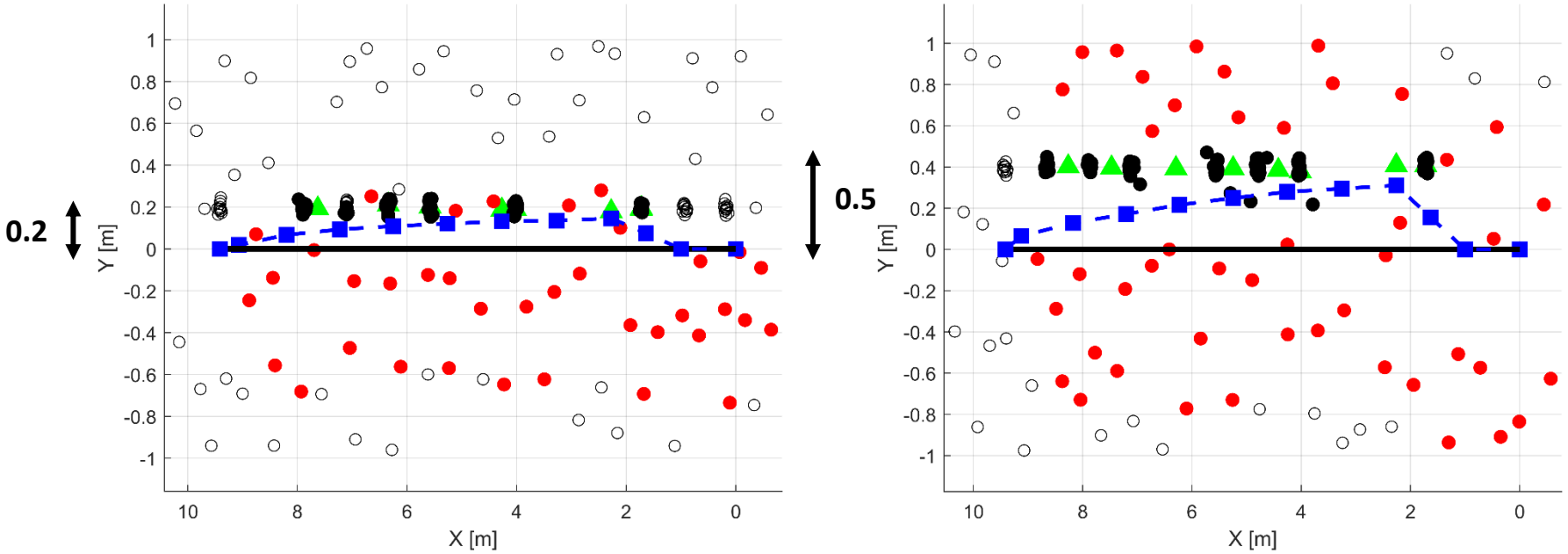




SIMULATION (2 / 3) - RIGHT

- Simulation environment
  - Drone FOV with a radius of 0.8 meters
  - Epsilon : 0.3, minimum points : 5
  - 474 feature points

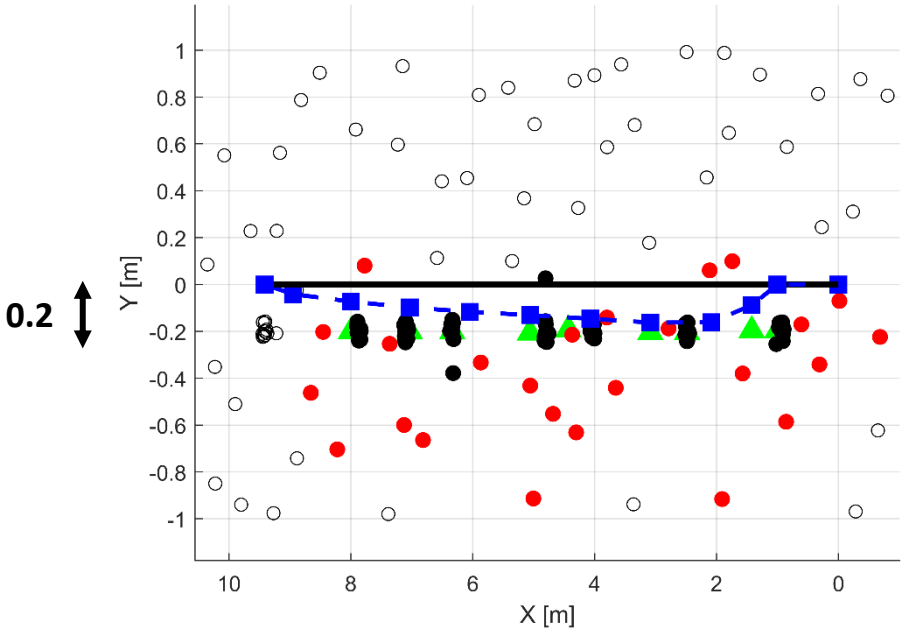
- Feature points considered in path planning
- Noise
- Optimal average point
- Feature points not considered in path planning
- LOS(Line of Sight)

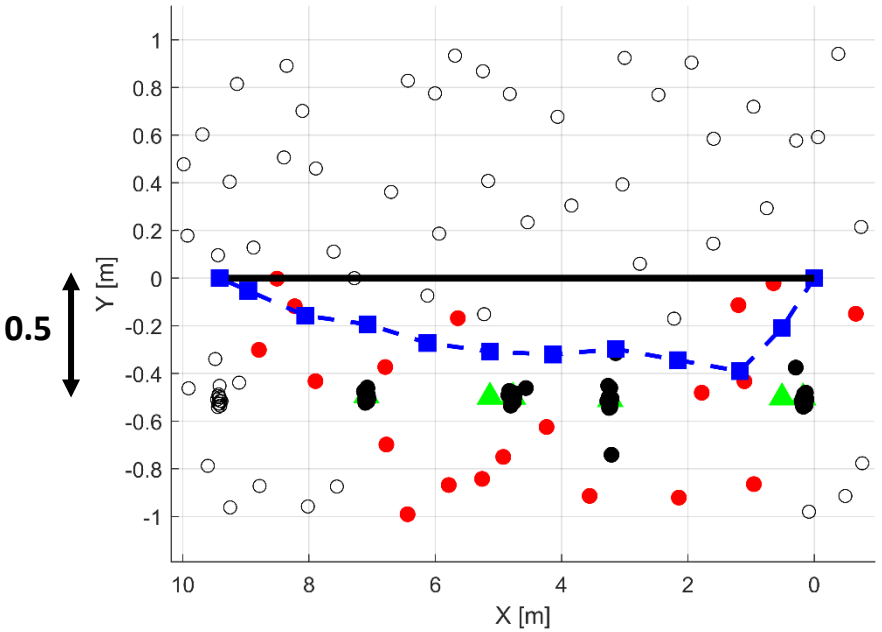


SIMULATION (3 / 3) - LEFT

- Simulation environment
  - Drone FOV with a radius of 0.8 meters
  - Epsilon : 0.3, minimum points : 5
  - 474 feature points

- Feature points considered in path planning
- Noise
- Optimal average point
- Feature points not considered in path planning
- LOS(Line of Sight)





CONCLUSION

- Feature based path planning algorithm
  - DBSCAN
    - Noise
    - Optimal average point
  - weight
    - The number of feature points
    - The density of feature points
  - waypoint
    - Feature point vector
    - Goal point vector

Q&A